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Reference Synthetic Dataset for Drilling Inventory Optimization

O. E. Abdelaziem, Amal Petroleum Company, Egypt; R. M. Khafagy, BP, Egypt; H. M. Darwish, SLB, Saudi Arabia; M. M. Abdelrahman, ADNOC Onshore, United Arab Emirates

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Summary

Data analytics holds immense importance in optimizing inventory management in the oilfield due to its significant business impact. However, the progress of research is limited by restrictions of sharing inventory data publicly, owing to privacy concerns. This paper introduces a novel synthetic dataset for drilling inventory systems in oil-integrated companies, which shall offer a valuable open-source reference for future studies. Additionally, diverse analytical approaches were employed to showcase their effectiveness in leveraging cost reductions.

Generating synthetic datasets involves the implementation of data-driven and process-driven methodologies. Given the inherent risk of endogeneity bias, a process-driven approach, combined with domain knowledge, was followed. Primarily, pre-defined criteria were established to govern dataset generation while accounting for various possible well templates. In essence, multiple constraints outlined the relationship among oil prices, expected annual drilling activities, timing and cost of item receipts, and the interdependence of item quantities and issuances. Next, controlled noise was introduced to account for irregularities and periodic inclusion of cutting-edge items. Ultimately, diverse inventory scenarios were subjected to analytics and optimization.

The introduced inventory dataset is characterized by randomness, rule-based procedure, and iterative process. It allows replicability and adaptability to divergent layouts by virtue of user-defined parameters. Overall, it possesses a history of transactions that extends over 20 years, exhibits an average asset value of 5 million USD, features around 500 unique stock keeping units (SKUs), and accommodates more than 15,000 transactions, such as receipts, issues, returns, sales, loans, inspections, and junks. Initial inventory volume was estimated based on the drilling activities of a hypothetical medium-sized oil company that operates an average of 10 drilling and workover wells per year. The generated dataset covers abundant specifications of multidisciplinary SKUs, covering tubular goods, drilling accessories, completion equipment, and wellhead materials. Items were selected delicately to represent diverse well configurations, encompassing sidetracks, slim holes, gravel-packed wells, artificially lifted wells, and dual producers. Furthermore, probable mistakes and human errors were taken into account, to improve the system's reliability to closely mimic real-world conditions. Subsequently, and upon investigating various Key Performance Indicators (KPIs), it was

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observed that analytics and clustering techniques are promising in elucidating latent challenges, maximizing profits, and achieving cost savings up to 10% of the inventory's total value.

This paper presents a novel, openly accessible synthetic dataset for drilling inventory systems in oil-integrated companies for the first time in literature, to the best of authors' knowledge. The introduced dataset was meticulously designed to ensure scalability and minimize drift. Positioned as a reference for future research, this dataset aims to drive forward the advancement of research in analytics and machine learning, to deliver ingenious solutions for inventory management optimization in digital oilfields.

Introduction

Inventory data is often restricted and unpublishable due to concerns about data privacy, competitive advantage, intellectual property, regulatory compliance, and data misinterpretation, negatively impacting research initiatives. Bellovin et al. (2019) demonstrated that synthetic data sets balance the tradeoff between utility and privacy. El Emam et al. (2020) elaborated three types of synthetic data, which could be generated from actual datasets, without using real data, or using a hybrid technique. Alnumay et al. (2022) revealed the importance of generating synthetic data sets in oil industry to develop AI solutions as an alternative to actual field data, especially when real data is restricted. Eldred et al. (2023) highlighted the significant, yet under-researched potential of artificial intelligence (AI) in inventory management in oilfields and discerned that data analytics and machine learning would be feasible only with appropriate dataset, which is defined as the one containing representative information on drilling sequence data, stock data, and consumption data.

This research has two objectives: 1) majorly, to create a synthetic reference dataset for drilling inventory without using real data, and 2) minorly, to show the role of analytics in inventory management and optimization. The data set was generated based on simulation runs, and multidisciplinary domain knowledge of supply chain in the oil industry along with drilling and completion engineering expertise, which ascertained the development of resilient inventory system aligned with real-data and matches actual drilling operations. This work shall serve as a benchmark for researchers to innovate and apply novel supply chain mathematical models to optimize the holding inventory value in upstream operating companies. The generated dataset is openly accessible, replicable, and available on https://github.com/OsamaEA/SPE_220950_MS.

This paper is organized as follows. In *the Introduction* section, earlier work on the dual objectives is demonstrated under *preceding studies*, followed by declaring the problem statement. The *Inventory Management Review* section summarizes supply chain and inventory control problems. The *Methodology* section details the workflow used to generate the synthetic dataset. In the *Results* section, a step-by-step approach is followed to explore how the generated dataset mimics real inventory data. Under the *Discussion* section, statistical insights and the role of analytics have been investigated on the introduced data set. In the section of *Limitations and Future work*, possibilities for ongoing research are outlined. Finally, the *Conclusion* section summarizes the key findings.

Preceding studies

It was reported that more than 89 articles in research have addressed SCM in oil and gas companies with over 52 distinct SCM models being investigated (Alhosani et al. 2019). Inventory management in the oil upstream industry faces numerous challenges, including unpredictable lead times, demand fluctuations, biased forecasting, pipeline stock, SKU proliferation, and SKU distribution mix error. The ultimate objective of research is to optimize the ratio of profitability.

Synthetic-Dataset-Based Inventory Analytics in Oil Upstream Industry. Based on a synthetic data set, Marcellino and Sichman (2010) proposed local optimization and multi-agent approaches for a supply chain, which integrates three terminals of two refineries and a drilling location. Montagna and Cafaro (2019) developed a framework to optimize SCM, using both a hypothetical illustrative scenario and a real case

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from an oil and gas company in South America, with the objective of minimizing costs of inventory holding, investments, and operations.

Inventory Analytics. Ali et al. (2020) examined inventory controlling efficiency of an oil producing company in terms of backorder and service levels. Their study, which included attributes of annual demand, per unit cost, and lead time in days, was limited to a data set of the top 10 class A items. Laskar (2022) showed that implementing analytics in smart warehouse management systems for onshore and offshore oil company locations enhanced the overall supply chain performance with seamless integration with an ERP system. Ahunanya et al. (2022) addressed inventory management systems in drilling fluid firms through a questionnaire and statistical assessments to evaluate the relationship between various business metrics and financial inefficiency in stock systems.

Synthetic Datasets Generation. Bellovin et al. (2019) defined synthetic data as a sort of fake data derived from real data, highlighting its value in mirroring real data distributions and being indistinguishable from it. Jordon et al. (2022) defined synthetic data sets as those created using mathematical models or algorithms, aiming at addressing data science problems. El Emam et al. (2020) introduced the concept of utility to measure how well a synthetic dataset serves as a proxy for real data, stating that analysis results from synthetic data should resemble outcomes from real data. They demonstrated that synthetic data sets could be created without real data, based on analyst's background knowledge, statistical models, or simulations. This background knowledge comes from textbooks, historical conditions, statistical distributions, and experience. Additionally, involving domain specialists in this process ensures that the synthetic data aligns with real data, enabling assumptions about distributions and correlations when historical data is unavailable.

Research Gap and Problem Statement

Drilling inventory expresses an expensive asset in oil integrating companies, yet detailed investigations into inventory KPIs are sparse, owing to limited availability of data for research purposes. Consequently, generating a synthetic dataset became inevitable to allow researchers to develop innovative inventory management approaches in oil operating companies, which might ultimately lead to identifying financial issues and achieve substantial potential savings.

The introduced novel approach addresses the generation of a large-scale synthetic data set of drilling inventory system of a medium-sized oil company, establishing a benchmark for future work on investigations of drilling inventory optimization. Inventory management metrics were also evaluated on the developed data set, to ensure its reliability. Data is open-source, codebase was developed using python, and the final output is presented in PDF, which replicates actual reports of inventory transactions in oil companies.

Motivation for Generating Synthetic Data. Goldfarb and Tucker (2012) declared that digitization has interconnected privacy policy with innovation policy, arguing that data privacy represents the tradeoff between data-driven innovations and potential risks of data sharing. Bellovin et al. (2019) revealed that perfect privacy is achieved when no data is shared, while optimum utility is achieved when data is not perturbed. Few early studies implemented hypothetical data to depict the effectiveness of synthetic data sets in drawing conclusions for SCM in the oil industry (Montagna and Cafaro, 2019; Marcellino and Sichman, 2010).

Review of Inventory Management

Stevens (1989) delineated supply chain (SC) as a series of interconnected activities concerned with material planning and information flow, extending to supplier and manufacturer management, purchasing, facilities planning, and customer service, with an ultimate objective of balancing the tradeoff between high customer satisfaction, wide product range in market, low financial working capital, low inventory investment, and low

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unit cost. Mentzer et al. (2001) defined SC as a network comprising at least three entities, either individuals or organizations. Houlihan (1988) differentiated the emerging concept of SCM from traditional materials control, mentioning that SCM integrates diverse perspectives to achieve joint strategic decision making, unlike classical SC, which was viewed as a single process with separate responsibilities.

In year 2015, American Production Inventory Control Society (APICS) defined inventory or stock as items used to support production (raw material and work-in-process), activities (maintenance, repair, and operating supplies), and customer service (finished goods and spare parts). Vrat (2014) identified inventory as the central theme of SCM, describing it as idle resources that tie-up financial capital without being productive. Aro-Gordon and Gupte (2016) demonstrated four types of inventories, known as slow-moving, fast-moving, dormant, and obsolete. Ivanov et al. (2021) outlined that inventory is typically an expensive asset, which could hold up to 50% of total invested capital of companies where the fundamental costs are outlined as carrying costs, ordering costs, and shortage costs. Naddor (1966) mentioned that inventory problems arise when a decision is needed to initiate a purchase order – determining when to buy – and set reorder points – determining how much to buy. Inventory should be addressed as a symptom of problems, which could impact the entire SC unless root causes are investigated, as naïve inventory reduction can lead to shortage of items and extra procurement costs.

Inventory Problems and Analysis

Wild (2017) pointed out that inventory could suffer from obsolescence, slow systems, and extra stock due to inefficiencies in determining accurate quantities of a purchase order (PO), identifying reliable safety stock, forecasting market changes, and poorly defined responsibilities. Shenoy and Rosas (2018) depicted the key issues of inventory as demand, replenishment lead time, inventory level, review time, lifetime, and repairability. Ivanov et al. (2021) mentioned that optimizing the trade-off between service levels and costs is the challenging core of inventory management.

Role of Analytics. Naddor (1966) stressed the necessity of inventory models, which establish mathematical relationships between holding costs, stockout, and ordering costs. Hausman (2001) demonstrated that analytics allow companies to measure tied funds and compare inventory levels across different categories. Kleijnen and Smits (2003) distinguished between theoretical SCM evaluation metrics and those used in practical business. Souza (2014) underscored the critical role of data analysis, exact algorithms, and approximate formulas in designing inventory policies, particularly forecast patterns, replenishment lead times, service levels, and holding costs. Kache and Seuring (2017) showed that the origin and type of information are fundamental dimensions for big data analytics in SCM. Nguyen et al. (2018) examined the role of big data analytics – volume, variety, velocity, veracity, and value – in optimizing inventory control dynamics, warehouse operations, order picking problems, and storage assignment.

Inventory Business Metrics. Gunasekaran et al. (2001) listed key performance metrics of inventory systems and classified them based on their financial impact. Those metrics could be categorized into three levels: 1) strategic, including total cash flow, return on investment, delivery lead time, cycle time, and customer query time, 2) tactical, encompassing demand and forecast, product cycle time, and transportation costs, and 3) operational, involving carrying costs, manufacturing costs, and capacity utilization. Hausman (2001) depicted two primary metrics for inventory evaluation: inventory valuation, representing total monetary worth of unsold materials, and inventory turns, which are connected with stock flow and time supply. He also highlighted the importance of speed metrics, such as cycle time and cash conversion cycle, for effective inventory monitoring.

Vrat (2004) delineated inventory policies as the standard operating procedures for inventory models, highlighting the common policies such as 1) economic order quantity (EOQ) – Reorder point (ROP) policy, 2) periodic review inventory policy, and 3) optional replenishment policy. Aro-Gordon and Gupte (2016) distinguished the need for precise calculations of diverse stock levels for effective inventory management,

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including re-order level, maximum level, maximum level of consumption, average stock level, danger level, economic order quantity, and ABC classification. Shenoy and Rosas (2018) discussed strategies for reviewing inventory systems such as fixed order quantity system and order-up-to-level system, whether continuous or periodic. Nallusamy (2021) presented a holistic inventory planning workflow, which extends to ABC-FSN analysis, demand forecasting, ordering policy, inventory cost, re-order point, safety stock, inventory record accuracy, turnover, and logistic performance metrics.

Methodology

Workflow began with scoping the objective of creating a synthetic dataset of drilling inventory and simulating real operations data associated with rigs. A systematic and replicable process (Fig. 1) was followed, comprising five main blocks: (1) drilling schedule blueprint, (2) comprehensive approach to inventory forecasting, (3) inventory framework, (4) simulation of rig orders, and (5) database validation.

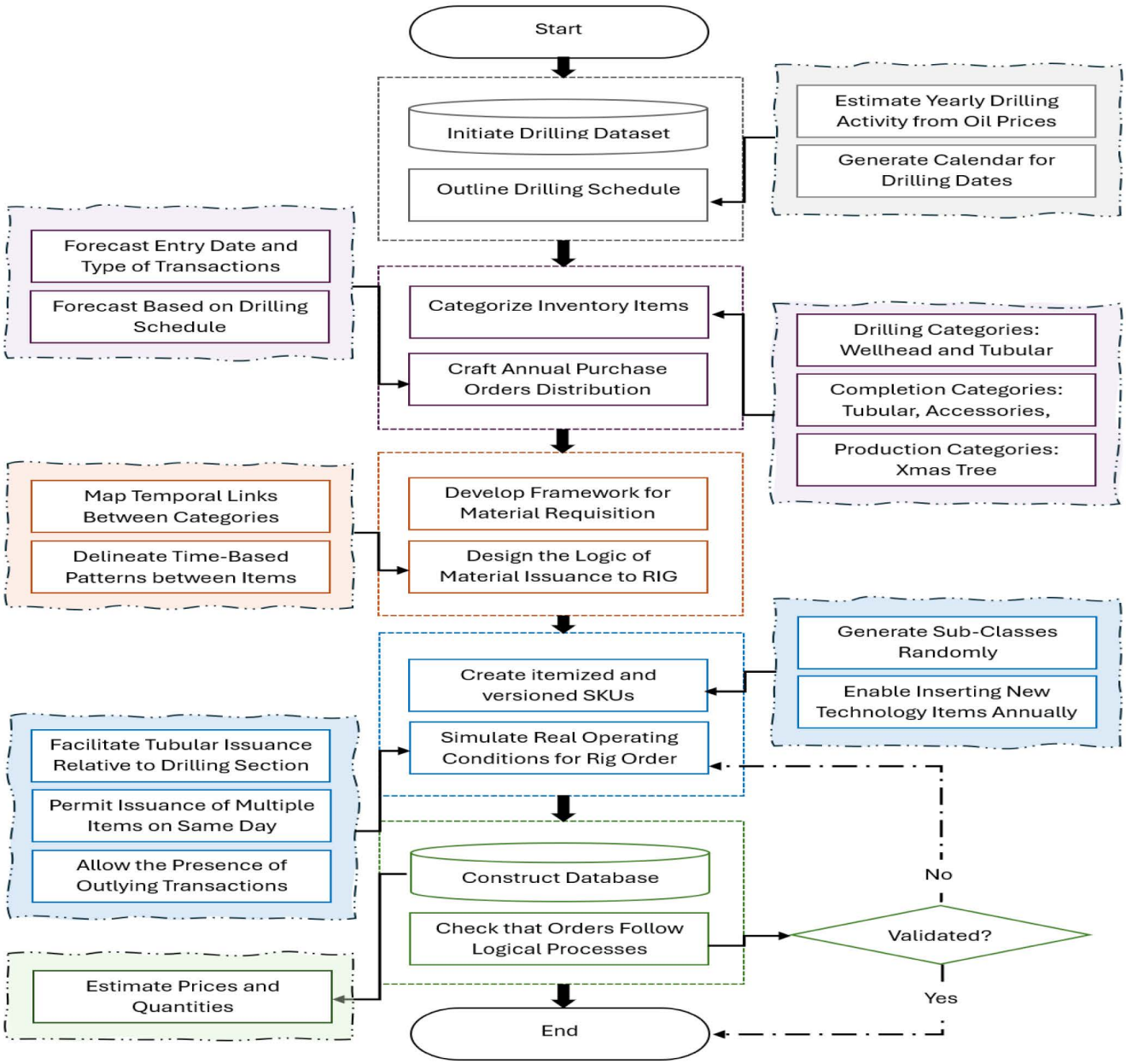

Figure 1—Summary of workflow for synthetic data set generation – consisting of five major blocks

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The first step focuses on initiating a drilling schedule for a hypothetical company and generating dates of annual drilling campaigns over a period of 20 years. The second step pertains to processing POs and receiving the requested equipment into inventory, aligning materials requisitions with drilling plans to mimic real oilfield operations. The third step frames a high-level design for the inventory system, emphasizing major categories as well as establishing a temporal connection between drilling operations and materials issuance to support operations accordingly. The fourth step reveals an iterative approach to generate the subclasses of each item, incorporating new high-tech items annually besides accounting for anomalous transactions. The final step validates, evaluates, and checks the logical consistency of the output data set.

Phase I: Drilling Schedule Blueprint

This stage serves as the foundation of the workflow, constructing a database for a hypothetical oil operating company, involved in drilling activities for over 20 years. Synthetic dataset was constructed on the premise that inventory data, purchase orders, and transactions history are linked to the number of wells drilled each year. Typically, there is a direct correlation between oil prices and the number of wells drilled. Nonetheless, other factors such as investment decisions, economic factors, cost of production, and regulatory environment must also be considered. In this study, the first attribute was solely considered to construct the drilling schedule spanning 20 years, resulting in a total of more than 200 generated synthetic drilled wells.

Initiate Drilling Database. The process started with correlating the annual number of wells drilled to the average yearly oil prices. It was observed that economic growth, supply chain disruptions, and commodity prices impacted the US dollar inflation rates. Consequently, inflation-adjusted oil prices (Fig. 2a) were introduced to accurately estimate yearly drilling activities, factoring inflation rate in oil prices over the past 20 years. It was evident that inflation rates, compared to the current status, varied drastically from 0.78 in the year 2000 to 0.049 in the year 2022.

As shown in (Fig. 2a), the inflation-adjusted oil prices in years 2006 and 2022 are almost typical, although the actual oil prices in both years differed by more than 42%. This underscores the significance of considering inflation rates in oil prices and its impact on the number of wells drilled annually. It could be noticed that the persistent pattern of the synthetic data closely mirrors the actual trend of wells drilled globally from the year 2004 to the year 2022 (Fig. 2b). For the developed medium-sized company, it was observed that the annual drilling activities (Fig. 2c) reached the lowest in the year 2003 with only 4 wells, while peaked in 2008 with 14 wells, followed by a drop, which was impacted by the economic crisis. Similarly, it was observed that the number of wells drilled in year 2020 dropped to 5 wells, adversely impacted by the COVID-19 pandemic.

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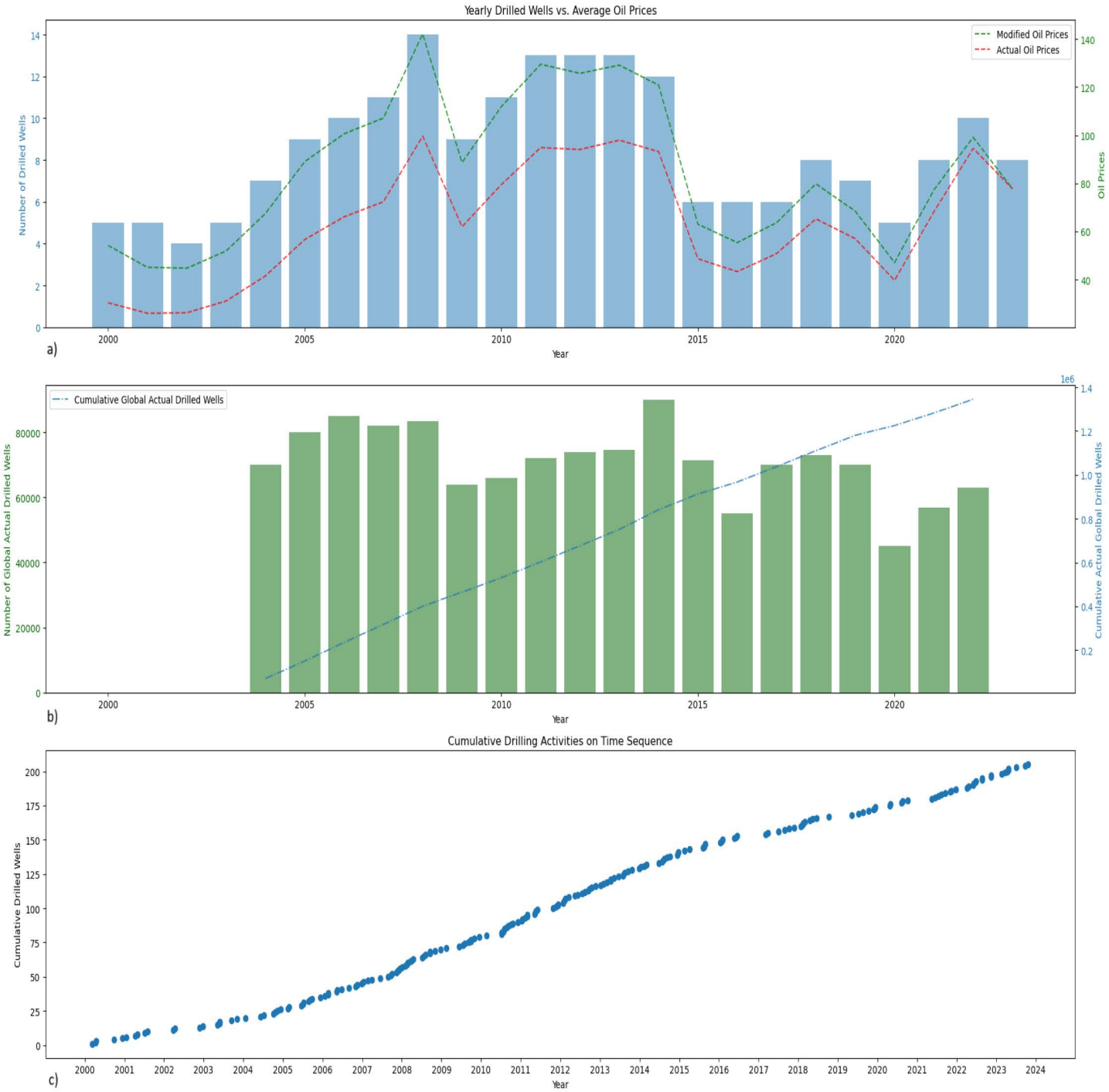

Figure 2—Drilling Blueprint Schedule: a) the annual drilled wells and inflation-adjusted oil prices for the synthetic dataset, b) the actual data of global drilled wells from 2004 to 2022 after (Bui and Luong, 2024), c) total number of wells drilled in the synthetic dataset.

Outline Drilling Schedule. In this stage, the dates of drilling activities, the duration of each campaign, and number of days that the rig remains spud on a well are estimated. This step is crucial as it significantly impacts the entire inventory workflow, owing to the fact that the transactions of materials requisition, issuances, returns, and purchase orders rely on sequential progress in drilling operations. To eliminate bias, the process of dates generation was conducted using a random and normally distributed approach, which resulted in the highest number of wells being drilled in the third quarter, totaling 59 wells, while the first quarter had the fewest drilled wells, totaling 49.

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Phase II: Holistic Approach for Inventory Forecasting

Purchase orders are typically planned no less than one year ahead, taking into account the planned drilling operations for the following year. This planning accounts for long lead times, shipping expenses, and the urgency of receiving materials. Therefore, there is generally a maximum one-year gap between the POs and the planned drilling (Fig. 3a). Oil companies strive to optimize the ordering process and reduce costs. In this phase, inventory items are sorted into six primary categories, with annual estimates made for the inventory entries of each category.

Figure 3—Holistic Approach for Inventory Forecasting: a) Distribution of number of annual drilled wells analogous to POs, b) Major six categories of inventory items, c) Annual distribution of stock entries by category, d) Monthly seasonal distribution of POs entries.

Categorize Inventory Items. This approach primarily addressed drilling inventory, typically involving material requisitions for the rig site. The drilling process begins with conductor hammering, followed by running casing and liners, progressing to wellhead sections, completion tubular and equipment, artificial lift installation, Xmas tree assembly, and finally putting the well onto production. Accordingly, the generated dataset reflected those main categories and broke down the inventory into six primary catalogues (Fig. 3b).

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It could be observed that the catalogue of completion equipment or accessories has the highest number of receipt transactions – almost double that of any other category–of the stock, due to the extensive variety of completion equipment, the potential of POs splitting among different vendors, and varying delivery time for items based on manufacturer availability. In contrast, tubular receipts usually arrive in batches

Craft Annual Purchase Orders Distribution. A conservative approach was adopted, which assumes that there should be at least one received order and at most six received orders of material requisitions every year. A set of probabilities was presumed for the main six categories, each with a 0.10 probability of reception, except for completion accessories, which were assigned a 0.50 probability, i.e. the likelihood of receiving completion equipment is equal to the combined probabilities of receiving all other material. This process was randomly generated, accounting for the possibility of materials being received in multiple shipments and the fact that some materials might not need to be purchased annually. Distribution of yearly POs and dates of materials entries into the inventory system (Fig. 3c) followed an iterative approach, which permitted complete randomness and considered the current status of the inventory system. X-mas tree, for example, experience 65 entries across the history of stock transactions, with an average of 3 entries per year, and a minimum of 1 entry achieved in five distinct years and a maximum of 6 entries achieved once in year 2012. Conversely, wellhead equipment achieved a maximum of 5 entries in years 2006 and 2007 only.

Furthermore, seasonal patterns of POs were observed in (Fig. 3d), which could be influenced by various factors, including adjustments to the drilling schedule, weather conditions – especially for offshore rigs affected by wind speed, wave height, and swell – market demand, regulatory requirements, and operational constraints

Phase III: Inventory Framework

At this stage, a broad outline of the inventory system was presented, highlighting the general transactions of issues and receipts that a drilling inventory might experience. Up to this point, inventory items are still being considered as six primary groups, without defining subclasses. This was deliberately done, to prevent any potential bias or inaccuracies that might arise from dealing with detailed inventory aspects.

Develop Framework for Material Requisition. The process started by defining a description for each main category followed by a suffix indicating the batch number issued to a single rig site. For instance, a drilled well might require multiple *n* deliveries of drilling tubulars to the site, based on well engineering design and the number of sections to be drilled. This would result in multiple issuances to the rig labeled as drilling-tubular#1, drilling-tubular#2, and so on, up to drilling-tubular#*n*. The adopted approach was applied across all the six main categories, with the maximum number of issuances and returns (Table-1) to each rig generated randomly according to constraints derived from real-world experience of drilling operations. This ensured that the developed system could realistically simulate rig conditions.

Table 1—Material Requisition Framework showing criteria for the design

Item	Category	Number of Issues per Well		Number of Returns per Well	
		Minimum	Maximum	Minimum	Maximum
1	Drilling Tubular (Conductor and Surface Casings)	0	3	random (0, max (1, *N* issues - 2))	*N* issues + 1
2	Drilling Tubular (Casings and Liner)	2	5		
3	Drilling Wellhead	*N* issues of drilling tubular			
4	Completion Tubular	1	2		
5	Completion Equipment and Accessories	1	3		
6	Artificial Lift Equipment	1	2		
7	Xmas Tree Equipment	1	2		

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As an illustrative example, it can be seen that the number of returns for each item per well follows a consistent pattern, where a value is randomly selected between a minimum and a maximum. The minimum is a random value between zero and the higher of one or (number of issues minus two), while the maximum is the number of issues plus one. For instance, for drilling tubulars, the number of times they are issued to a well would be a randomly generated number between 2 and 5, assuming it is 4. For the same item, the number of returns would be a randomly generated value between a random value between 0 and 2 against 5. If this random value is 2, then for this well, there would be issuances of drilling-tubular#1, drilling-tubular#2, drilling-tubular#3, drilling-tubular#4, and returns of drilling-tubular#1 and drilling-tubular#2.

Statistical insights from issue and return transactions are depicted in (Table-2). It is clear that drilling tubular are the most frequently issued items to rig, while completion tubulars are the least. This could be attributed to the fact that operational problems typically occur during the drilling phase as opposed to the completion phase. Such problems might result in changes in well planning, non-productive time, and other problems, which could adversely impact the material transactions to and from the rig. It is obvious that the trend in issuances of most categories aligns with their return transactions, yet with slightly lower means, indicating that more items are issued than returned.

Table 2—Statistical Insights from Inventory Issues and Return Transactions

		Issue Transactions						Return Transactions					
Item	**Category**	**Count**	**Mean**	**STD**	**Q1**	**Median**	**Q3**	**Count**	**Mean**	**STD**	**Q1**	**Median**	**Q3**
1	Drilling Tubular (Conductor and Surface Casings)	*392*	*1.63*	*0.73*	*1*	*1*	*2*	*296*	*1.36*	*0.58*	*1*	*1*	*2*
2	Drilling Tubular (Casings and Liner)	*715*	*2.43*	*1.25*	*1*	*2*	*3*	*524*	*2.09*	*1.12*	*1*	*2*	*3*
3	Drilling Wellhead	*715*	*2.43*	*1.25*	*1*	*2*	*3*	*539*	*2.13*	*1.16*	*1*	*2*	*3*
4	Completion Tubular	*295*	*1.31*	*0.46*	*1*	*1*	*2*	*249*	*1.18*	*0.38*	*1*	*1*	*1*
5	Completion Equipment and Accessories	*420*	*1.69*	*0.76*	*1*	*2*	*2*	*308*	*1.41*	*0.63*	*1*	*1*	*2*
6	Artificial Lift Equipment	*210*	*1.22*	*0.39*	*1*	*2*	*2*	*180*	*1.03*	*0.26*	*1*	*1*	*2*
7	Xmas Tree Equipment	*301*	*1.32*	*0.47*	*1*	*1*	*2*	*247*	*1.17*	*0.37*	*1*	*1*	*2*

Design the Logic of Material Issuance to RIG. Overall, this step creates a logic that synchronizes inventory transactions with operations at rig site. Referring to the six main categories of drilling inventory, each issuance to the rig would be temporally correlated to the spud-in date, besides being linked to the issuance of other material to rig site. For example, drilling tubular of conductor would be the first equipment to be issued to the rig site. In some cases, they might not be issued, assuming that the well is sidetracked and already have conductor and surface casings in place. Similarly, casing size of 7" should only be issued to the rig provided that casing size of 9 5/8" has already been issued or provided that it is a sidetrack in a well that was originally designed with 9 5/8" casing. Likewise, different sizes of drilling wellheads should be issued along with drilling tubular. It is important to notice that the introduced logic addresses drilling wells solely, and that it does not apply to workover wells, which would neglect items#1 to #3.

Time-based sequence of inventory SKUs transactions was designed based on practical experience and well engineering principles (Table-3). This process was applied to both issues and returns for each item. First items to be issued to the rig are conductors and surface casings, which are assumed to happen within two weeks from rig spud-in on a well. In contrast, wellheads are temporally associated with the issuance of drilling tubulars, such that they typically occur within a range of one week before or one week after the issuance of drilling tubular. For returns to the inventory system, it was assumed that SKUs, whether new,

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used, or to be junked, would return to stock within two to twelve weeks, accounting for potential delays due to re-inspection and transportation. This methodology also considered that some items might be sent to the rig and wait on weather or repair. The process was entirely random to eliminate subjectivity.

Table 3—Temporal Logic of Inventory Items showing correlations between items issuances and returns

Item	Category	Time-Based Reference	Number of Days to Issue from Reference Base		Number of Days to Return from Reference Base	
			Minimum	Maximum	Minimum	Maximum
1	Drilling Tubular (Conductor and Surface Casings)	Rig Spud-in	0	14	15	90
2	Drilling Tubular (Casings and Liner)	Previous Item	15	90		
3	Drilling Wellhead	Previous Item	−7	7		
4	Completion Tubular	Item#2	3	14		
5	Completion Equipment and Accessories	Previous Item	−3	10		
6	Artificial Lift Equipment	Item#4	−3	10		
7	Xmas Tree Equipment	Item#4	−3	5		

Referring all dates to the spudding date, distribution of materials issuance and return for each category is statistically demonstrated in (Fig. 4). It could be seen that distribution patterns vary considerably among categories. In rare exceptional cases of offshore wells with extended drilling schedules, the entire materials of a single well could be issued within a maximum of 4 months and returned within a cap of six months. Statistically, conductors are issued on average of 5 days from rig spud-in. Similarly, drilling tubulars typically reach the rig site and cover drilling operations within an average period of 60 days. Conversely, completion material, which are moved in the final phase of well completion, are typically handled within 90 days on average, ranging from a minimum of 33 days to a maximum of 122 days.

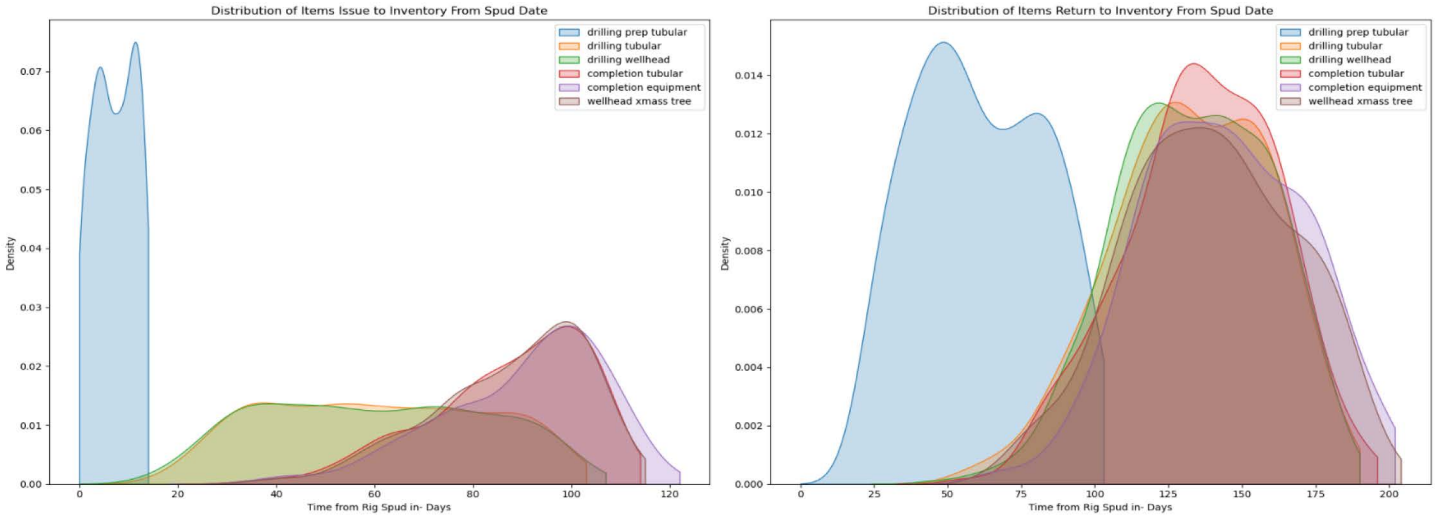

Figure 4—Timeframe from rig spud-in to materials issuance and return by category

Phase V: Simulating Rig Orders

The core task of synthetic data generation belongs to this step, in which subclasses for each of the six primary categories are randomly created. Moreover, this iterative process accommodates the regular addition of high-

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tech items into inventory and permits the presence of obsolete items. In addition, it incorporates exceptional transactions such as inspection, loan, sales, and junk.

Create Customized and Multi-Versioned SKUs. This section centers on the dates of entries and the description of SKUs. Each category is divided into subclasses, reflecting items with different technical specifications, manufacturers, or varying conditions of use. Constraints were established, so that each unique item in the inventory shall be broken down into a minimum of 5 subclasses and a maximum of 10 subclasses, following a left-skewed exponential distribution. More constraints were applied to replicate real operations and maintain the introduced logic. According to this initial setup, the original stock items were defined, and their entry dates were projected. It was assumed that some items were already part of the inventory in the year 2000, with no information about their earlier history of transactions. Therefore, it was presumed that the entry date for those items was the first day in the year 2000 and random quantities for each item were assigned.

The objective of the introduced constraints was to guarantee that a subclass item cannot be issued before it had been already purchased and received through a PO. Similarly, any item returned to the stock must have been issued to a rig earlier. In addition, an iterative criterion ensured the possible presence of obsolete SKUs, where each subclassed item was associated to a skewed probability of 0.15 of being obsolete in a given year. This approach simulates real scenarios in inventories, where an item is assumed obsolete of the company decided to discontinue their use in operations for any reason.

In order to integrate high-tech or new specifications of items in the inventory system, the period from 2000 to 2023 was split into four groups, with splitters randomly selected, such that each segment could cover a duration ranging between 5 and 9 years. For each segment, a random number of the newly added items was estimated based on a normal distribution. This iterative process was applied to each subclassed item. For a straightforward example, assume the first splitter is a value randomly chosen from years between 2002 to 2004 and the second splitter is selected randomly from year 2007 to year 2009. The generated segment could be supposedly picked from 2003 to 2009, covering a span of 6 years. Based on this segment, the number of new items designated to a unique class would be randomly determined to be between 2 and 4.

This reproducible approach resulted in the generation of 500 unique SKUs for drilling inventory, distributed across diverse categories with unique number of entry dates (Table-4).

Table 4—Distribution of unique subclassed items

Item	Category	No. of Unique Items	No. of Unique Entry Dates
1	Drilling Tubular	100	58
2	Drilling Wellhead	55	57
3	Completion Tubular	50	54
4	Completion Equipment and Accessories	190	117
5	Artificial Lift Equipment	47	57
6	Xmas Tree Equipment	61	65

Moreover, it could be observed that the distribution of the first entry dates of items in the inventory and the last entry dates of items in the stock are left and right – skewed respectively (Fig. 5a). Furthermore, the average inventory retention time, which represents how long SKUs remain in the stock before being issued, is depicted in (Fig. 5b). In simple terms, for an SKU, the first entry date refers to when the item was first received from a purchase order, while the last entry date is indicating the most recent time an item had been added to the stock. It could be viewed that several items have their last entry date earlier than 2020, suggesting that they are no longer in demand by the company, indicating potential obsolescence. Similarly, the majority of first entry dates correspond to the year 2000, which refers to items that have already been

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part of the stock before this date. Similarly, the average retention time is typically less than 18 months for most items.

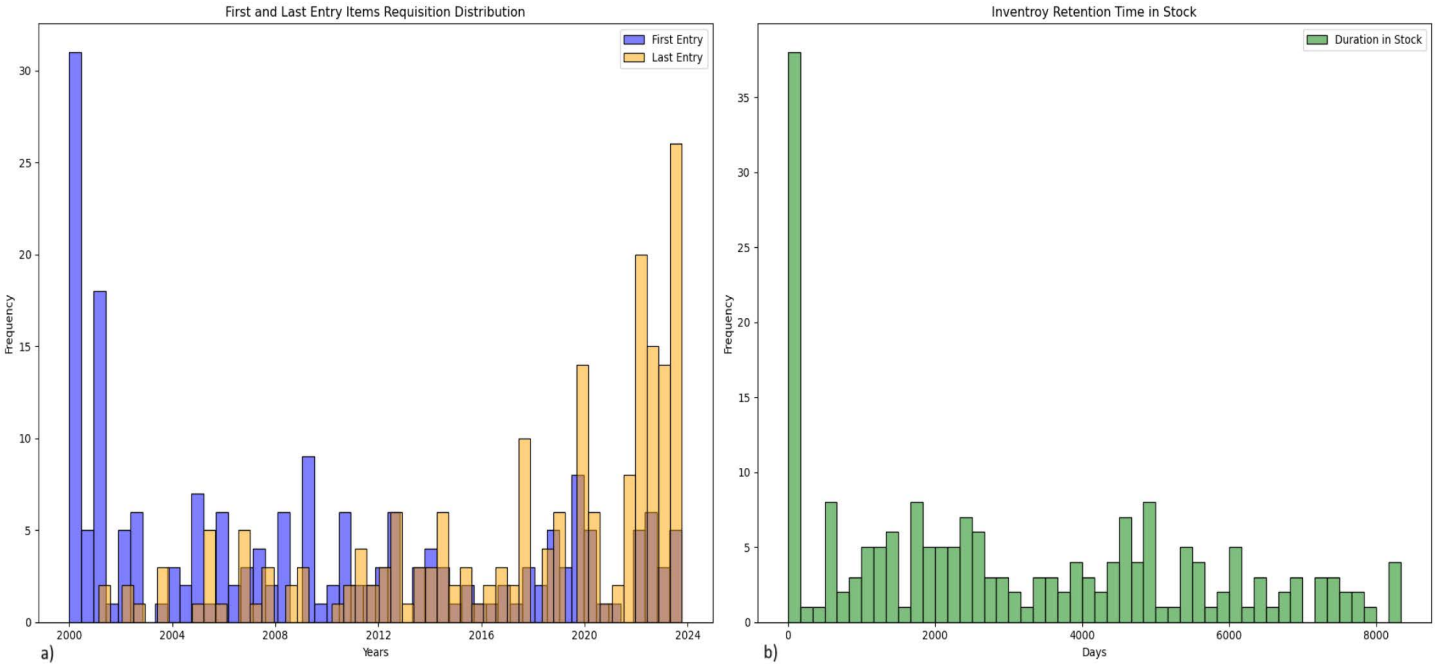

Figure 5—Entry transactions of purchased items: a) distribution of first and last entry dates for SKUs, b) histogram of retention time of SKUs

Simulate Real Operating Conditions for Rig Order Issuance. In this section, the framework of all potential inventory transactions was outlined, including routine transactions of issues and returns, as well as atypical ones such as junk, sale, loan, and adjustments (Table-5). Sequence of SKUs orders, relative to the current drilling hole section was examined. Possibility for issuing multiple SKUs at the same time to the rig site or issuing SKUs irrelated to the current drilling phase were also addressed, compensating for offshore conditions and remote locations.

Table 5—Definitions of inventory transactions

#	Transaction	Definition
1	Receipt	The addition of items into inventory, typically as a result of receiving goods from suppliers, returns from customers, or completed production items.
2	Issue	The removal of items from inventory to be used in operations, production, or for fulfilling orders.
3	Return	The process of receiving items back into inventory that were previously issued to operations.
4	Adjustment Quantity	The quantity of items that have been adjusted in the inventory, either added or subtracted, due to various reasons such as errors, corrections, or changes in inventory records.
5	Adjustment Price	The price associated with the inventory adjustments, reflecting the value changes due to discrepancies, corrections, or modifications in the inventory records.
6	Sale	The transaction in which items from inventory are sold to other companies. This type of sale involves business-to-business (B2B) transactions.
7	Loan Exit	The process of temporarily removing items from inventory to be lent out to another entity, with the expectation that these items will be returned at a later date.
8	Loan Return	The process of receiving items back into inventory that were previously lent out through a loan exit.
9	Junk	Items that are deemed unusable, defective, or obsolete and are therefore removed from inventory, often without any value recovery.

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Starting with conductor and surface casing category, the proposed design allowed tubulars to be issued to the rig as a batch that might consist of different casing sizes assuming that the rig had enough space to accommodate them. Given the short timeframe in running the surface casings, various combinations were considered. This framework also included the likelihood of sending the same casing size in multi-batches with a probability of 0.15, accounting for potential operational challenges. Similarly, the same logic was applied to drilling tubular of intermediate and production casings, completion tubular, accessories, and wellheads. This iterative process was crucial for maintaining scalability and sustainability of inventory transactions. It verified the size compatibility of drilling tubulars with tubing tubulars, for example, a 5" liner cannot accommodate a 4" tubing. Likewise, it confirmed that the sizes of completion accessories with tubing and casing tubulars are matching, such as a 7" × 3 ½" permanent or retrievable packer not being used with other tubular sizes in the well. This process was meticulously designed to guarantee that transactions adhered to the established logic.

Furthermore, this design considered dual producers, gravel pack, and artificial lift designs. Also, equipment that is usually received on a consignment basis has been included in this approach. Anomalous transactions were designed to occur rarely in the inventory with a 0.30 probability for any item. Items are assumed to have a possibility of being junked if 1) they had not experienced any type of transactions over N years, where N is a randomly generated number following a normal distribution mean of 10 years and a standard deviation of 5 years, or 2) damaged while movement, with a 0.25 probability for less than 0.15 of the returned equipment from the rig. In addition, loan transactions were designed with a criterion that loaned items should return to the inventory within a randomly selected period between 3 and 12 months. Furthermore, used items that return to inventory were classified into condition-B – items that are functional but might have minor defects from being used at most once during the operation – and condition-C – items that are still functional after being used multiple times. Items that are sold to other companies are priced based on the latest value of the same category in the stock.

Phase IV: Database Validation

At this stage, the data set has already been constructed successfully. This step introduces extra inventory data and reviews the workflow of the synthetic dataset and ensures meeting predefined conditions and criteria.

Construct Database. In this step, the quantities of each SKU received annually were designed according to the drilling campaigns planned in the following year. In contrast, the prices of each SKU were determined based on the current market knowledge of the drilling SKUs prices. Criteria that account for US dollar inflation rate, depreciation, and supply chain impacts were considered to estimate the historical prices of SKUs, when they initially entered the inventory system. In addition, Price fluctuations were also accommodated into the inventory tables.

SKU identifiers have been assigned to each unique item in the stock. Transaction number and requisition orders have been associated to SKUs. Analytical accounts responsible for transactions were added. In addition, columns of balance and value have been added. The database was organized in such a way that each item could be extracted into a separate table, providing a summary of each single transaction that it underwent. Overview of item technical description, item criticality, and warehouse locations were also introduced for each item. Data could be extracted either in PDF or Excel formats to facilitate further analysis.

Check that Orders Follow Logical Processes. To ensure seamless transactions, constraints are checked systematically throughout the process. The ratio between number of issues for dependent SKUs was validated to be within +/-0.10, and the receipt-to-issue ratio was set to be within an acceptable range between 1:3 and 1:8 (Table-6). For instance, 9 5/8" casing issues should closely match to 13 3/8" casing issues, given that both are likely used in subsequent section in drilling operations. Similarly, 5" casing issues should be less than 7" casings because the former would not run in a well unless the latter was already in place. In addition, 7" casings could be issued more than 9 5/8" casings, accounting for sidetracked wells that already

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have intermediate casings installed. It was also presumed that any item in inventory should be issued within a maximum of 5 years from its last entry date, however, exceptions were allowed with a 0.20 probability.

Table 6—Relationship between receipts, issues, and returns of tubulars

Item	Main Description	No. of Receipts	No. of Issues	No. of Returns
1	30" Conductor	28	75	75
2	24" Conductor	28	42	33
3	20" Casing	26	51	29
4	18 5/8" Casing	27	77	37
5	13 3/8" Casing	38	129	126
6	9 5/8" Casing	37	135	117
7	7" Liner	43	158	98
8	5" Liner	37	116	40
9	4" Tubing	40	139	117
10	3 ½" Tubing	32	71	62
11	2 7/8" Tubing	29	68	54

Results

In this section, the results from the methodology are presented, displaying the output dataset constructed at each step. This section validates the robustness of the introduced synthetic dataset, showcasing how it accurately mimics real field operations data.

Progressive Outputs Derived from Methodology

In this section, an illustrative example is introduced, depicting the expected output dataset from each block in the methodology. (Table-7a) shows the output schedule of drilling activities for the hypothetical company. (Table-7b) outlines purchase orders and reception transactions. In (Table-7c), issue and return transactions are generated and correlated to the drilling dates. (Table-7d) shows the combined dataset that includes diverse transaction types including anomalies. (Table-7e) demonstrates the final constructed dataset for a randomly selected SKU.

Table 7a—Progressive output of "Phase I: Drilling Schedule Blueprint" – Sample Example

Well No.	Well#1	Well#2	Well#3	Well#50	Well#100	Well#150	Well#200	Well#205
Drilling Date	12/03/2000	10/04/2000	27/09/2000	01/10/2007	10/08/2012	08/06/2016	23/04/2023	22/10/2023

Table 7b—Progressive Output of "Phase II: Holistic Approach for Inventory Forecasting" – Sample Example

Materials Requisition	Drilling Tubular	Completion Tubular	Completion Equipment	Artificial Lift Equipment	Wellhead Equipment	Xmas Tree Equipment
PO Reception	15/04/2000	21/01/2007	06/05/2014	22/09/2019	19/02/2022	08/01/2023

Table 7c—Progressive Output of "Phase III: Inventory Framework" – Sample Example

#	Transaction Date	Assigned on Well No.	Spud-in Date	Transaction Type	Transaction Category	Time from Spud-in
1	15/03/2000	Well#1	12/03/2000	Issue	Drilling Tubular (Conductor and Surface Casing)	3 days
2	30/04/2000	Well#1	12/03/2000	Return	Drilling Tubular (Conductor and Surface Casing)	49 days

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#	Transaction Date	Assigned on Well No.	Spud-in Date	Transaction Type	Transaction Category	Time from Spud-in
3	04/06/2000	Well#1	12/03/2000	Issue	Completion Tubular	84 days
…	…	…	…	…	…	…
5000	28/11/2013	Well#124	02/08/2013	Issue	Drilling Tubular	118 days
7000	08/08/2018	Well#168	18/04/2018	Issue	Completion Equipment	112 days
9000	05/06/2024	Well#205	22/12/2023	Return	Xmas Tree	135 days

Table 7d—Progressive Output of "Phase V: Simulating Rig Orders" – Sample Example

#	Transaction Date	Transaction Type	Main Category	Top Class	Subclassed Description	#Well
1	15/03/2000	Issue	Drilling Tubular (Conductor and Surface Casing)	Conductor	Conductor – Spec#3	Well#1
2	15/03/2000	Issue	Drilling Tubular (Conductor and Surface Casing)	Casing	20" Casing – Spec#1	Well#1
3	30/04/2000	Return	Drilling Tubular (Conductor and Surface Casing)	Casing	20" Casing – Spec#1	Well#1
4	04/06/2000	Issue	Completion Tubular	Tubing	3 ½" Tubing – Spec#2	Well#1
…	…	…	…	…	…	
1000	06/08/2005	Reception	Completion Equipment	Downhole Accessories	Dual Packer – Spec#1	NA
5000	26/03/2010	Sale	Artificial Lift Equipment	Gas Lift	Side Pocket Mandrel – Spec#2	Well#91
7000	06/08/2014	Junk	Xmas Tree Equipment	Valves	Wing Valve – Spec#1	Well#136
9000	17/11/2018	Loan – Exit	Completion Tubular	Tubing	2 7/8" Tubing – Spec#4	Well#164
15000	01/02/2024	Reception	Completion Equipment	Gravel Pack	4" Screens – Spec#6	NA

Table 7e—Progressive Output of "Phase IV: Database Validity" – Sample Example

	SKU Identifier: CSG-6832-TXZ	Main Category: Drilling Tubular	Top Class: Casing	Subclassed Description: 5" Casing, 15 lbs./ft, N-80, BTC connection		Part Number: AX-114386	Unit: PC	Last Price: 854 $	Warehouse: WH-01
#	**Transaction Number**	**Order / Requisition**	**Transaction Date**	**Transaction Type**	**Quantity**	**Balance**	**Price**	**Value**	**Analytical Account**
1		RC1578	23/07/2010	Reception	200	200	653$	130,600$	User01
2	I104678		24/09/2010	Issue	83	117	653$	76,401$	User05
3	R103845		30/11/2010	Return	17	134	653$	87,502$	User01
4	I105576		15/12/2010	Issue	111	23	653$	15,019$	User01
5		RC2699	31/05/2011	Reception	180	203	680$	138,040$	User03
6	I122486		10/02/2012	Issue	54	149	680$	101,320$	User05
7	I123586		07/04/2012	Issue	43	106	680$	72,080$	User05
8	I123588		10/04/2012	Issue	67	39	680$	26,520$	User02
9	R120286		22/05/2012	Return	31	70	680$	47,600$	User01
10	I124001		12/06/2012	Issue	46	24	680$	16,320$	User04
11	JK347		28/07/2012	Junk	3	24	680$	16,320$	User01
12	R120384		29/07/2012	Return	43	67	680$	45,560$	User02

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	SKU Identifier: CSG-6832-TXZ	Main Category: Drilling Tubular	Top Class: Casing	Subclassed Description: 5" Casing, 15 lbs./ft, N-80, BTC connection		Part Number: AX-114386	Unit: PC	Last Price: 854 $	Warehouse: WH-01
#	Transaction Number	Order / Requisition	Transaction Date	Transaction Type	Quantity	Balance	Price	Value	Analytical Account
13	I130674		11/01/2013	Issue	67	0	680$	0$	User01
14	R131463		05/03/2013	Return	7	7	680$	4,760$	User01
15		RC3158	29/10/2013	Reception	250	257	716$	184,012$	User01
16	I132499		04/11/2013	Issue	98	159	716$	113,844$	User05
17	R141687		06/04/2014	Return	9	168	716$	120,288$	User01
18	I142964		29/06/2014	Issue	87	81	716$	57,996$	User02
19	R150983		23/03/2015	Return	5	86	716$	61,576$	User02
20	I153687		20/06/2015	Issue	86	0	716$	0$	User01
21		RC5024	21/04/2016	Reception	70	70	906$	63,420$	User03
22		RC5024	22/12/2016	Reception	50	120	854$	102,480$	User01
23	I206752		29/10/2020	Issue	67	53	854$	45,262$	User02
24	R210076		15/01/2021	Return	2	55	854$	46,970$	User15
25	S210009		24/04/2021	Sale	35	20	854$	17,080$	User12

Final Database. Similar to the sample presented in (Table-7e), each of the 500 unique SKUs is presented in separate tables, which are then combined into a single or multi accessible PDF files. The primary objective was to create a database that resembles real drilling inventory databases, encompassing all potential transactions while accounting for the possible human and system errors in data recording. The provided example demonstrates an SKU that has been purchased for the first time to the inventory system in 2010. Extensive operations have been carried until 2015, resulting in more than 20 various transactions as stated. The reception-to-issue ratio was 1:3, which could be reasonable given the high demand for the item. It could be observed that multiple issuances have been transacted to the system, while a single return transaction was sufficient to accommodate them all, due to the possibility that there might be no return from certain rigs, when the items had been issued precisely as per the well engineering design without any back-up. In addition, inspection for the returned items has been factored in for through the allowance of junk transaction, which involves scrapping some items that returned from Rig in a damaged status. In the year 2016, it could be observed that there was a batch of the items, delivered on two consecutive dates with a six-month delay accounting for potential manufacturing and shipment problems. Similarly, it is apparent that there were no active transactions for the item in subsequent years, except for one issuance in 2020, indicating a possible flaw in the demand planning for the inventory system. This is corroborated by the sale of a large quantity later in year 2021. Following this, the item did not experience any further transaction despite being in the stock, suggesting that this module of this SKU might be outdated or obsolete according to the company standards.

Discussion

In this section, some insights from the final output dataset are presented. In addition, the role of data analytics in identifying pain points, particularly in major transactions of issues, returns, and receipts, was emphasized. In (Fig.6), timeseries analysis is showcased with (Fig. 6a) classifying the annual transactions by category, where seasonality could be viewed as expected, affected by oil prices and demand. To further demonstrate, a monthly timeseries of each transaction type was introduced in (Fig. 6b), highlighting significant seasonality

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across all transaction types. Accordingly, it was decided to perform a decomposition of the timeseries, however, the timeseries was resampled on a quarterly basis (Fig. 6c) due to data complexity. Decomposition separated the data into trend, seasonality, and residuals where trend is the long-term movement showing the overall increase or decrease in inventory levels, seasonality reflects the regular repeated patterns due to demand cycles, and residuals represent irregular variations or random fluctuations in inventory data. It could be clearly observed that issue and return transactions follow a similar pattern with shifted mean levels, whereas receipt transactions follow the same pattern with a one-year lag. Furthermore, it was noticed that reception transactions exhibit peak positive seasonality while issues display extreme negative seasonality values.

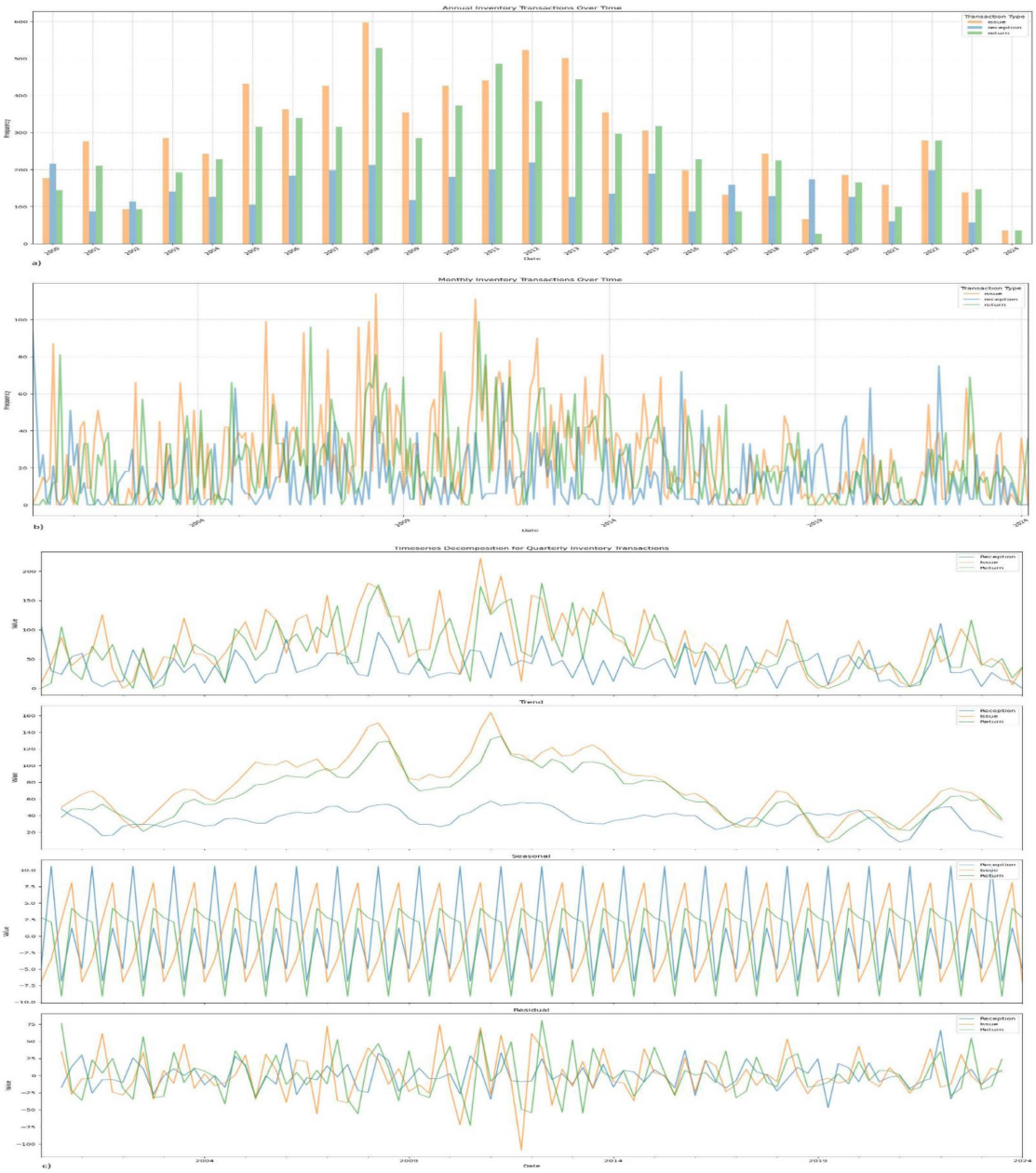

Figure 6—Time series analysis of inventory transactions over time as a) annual transactions of inventory categorized by transaction type, b) detailed transactions of inventory on monthly basis for each transaction type, c) time series complete analysis based on quarterly transactions showing the observed data, decomposed trend, decomposed seasonality, and decomposed residuals.

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(Fig. 7) depicts the distribution of the number of transactions typically experienced by an SKU. In this section, the main transactions of reception, issue, and return were considered only. To provide further details, (Table-8) demonstrates the statistical properties of these transactions. It is noteworthy that issue and return transactions have a mean value nearly twice as high as that of receipts.

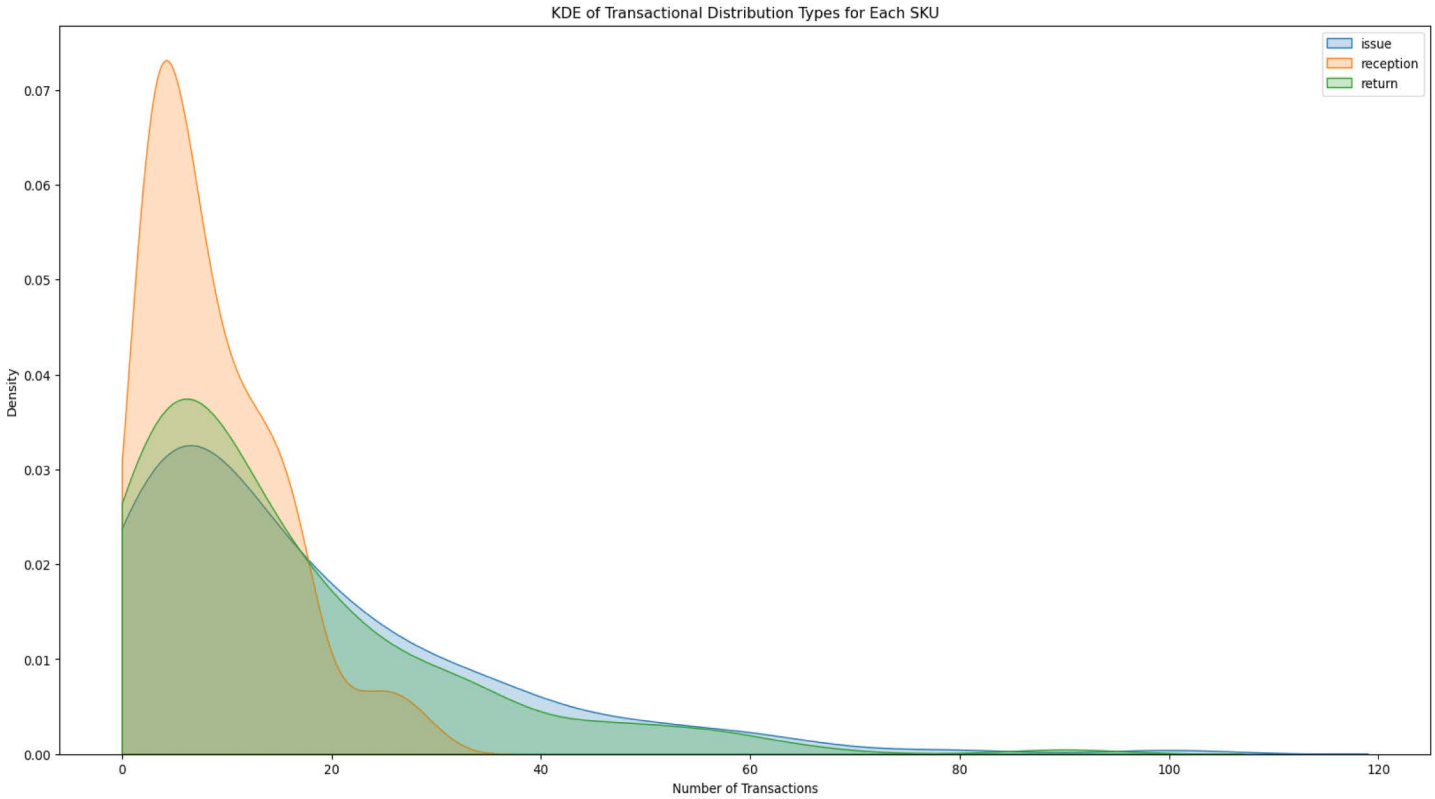

Figure 7—Distribution of Transactions for SKUs

Table 8—Statistical Insights from Transaction of SKUs

Transaction	SKUs Count	Mean	STD	Min	Q1	Median	Q3	Max
Reception	499	8.77	6.44	2	3.5	7	12	30
Issue	497	16.78	17.20	1	4	11	23.5	100
Return	490	14.91	15.37	1	4	10	19	90

In (Fig. 8), a demonstration of the issues-to-receipts ratio was introduced. It measures the balance between the outgoing and incoming SKUs, where a high ratio indicates stock depletion, with more items are issued than received, while a low ratio indicates stock accumulation. An optimum value of one suggests an efficient inventory management system. (Fig. 8a) displays the addressed ratio across the entire inventory system on a monthly basis, where strong variations are observed based on the accuracy of demand planning. For instance, the peak point in early 2008 likely indicates stock depletion subject to the high oil prices in that period, leading to increased drilling operations. Conversely, the low ratios in late 2008 indicates possible items accumulation, as the global financial crisis hit, reducing demand and creating uncertainty. Similarly, the same concept was applied to a sample of SKUs independently (Fig. 8b), showing that issue-to-reception ranged between 3 and 8, with some significant anomalies.

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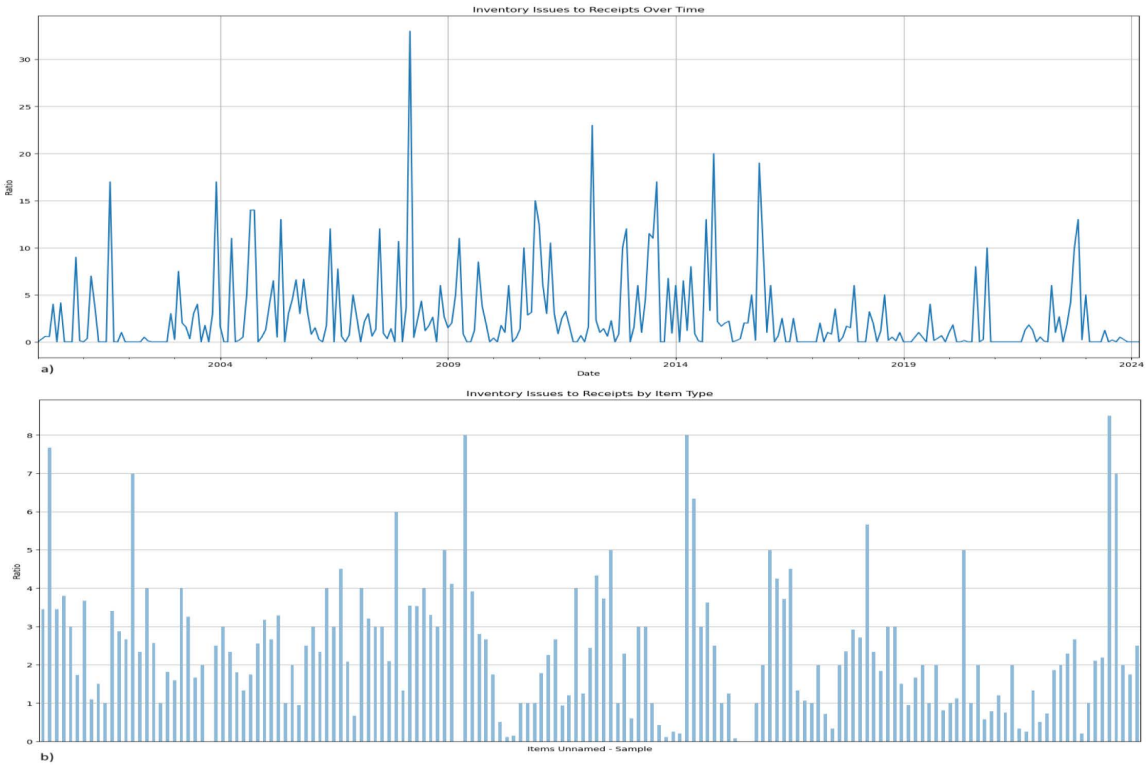

Figure 8—Issues-to-receipts where a) demonstrates the ratio for the overall inventory system, and b) categorizes the ratio per item as each bar represent the ratio for a different item from a random sample.

An overview of the inventory status is shown in (Fig. 9). A sample of SKUs was tracked as items enter, move through, and exit the inventory system. This sample was chosen to be 3 ½" tubing with six different technical specifications. It could be observed that some SKUs such as #spec-7 tubing only started to appear in the stock after 2014. Conversely, #spec-3 tubing did not experience any transactions after the year 2015, which indicates possible obsolescence and suggesting that this item has probably been replaced by #spec-7 tubing. Similarly, #spec-2 tubing experienced extensive transactions in the stock starting from year 2000 to year 2023, suggesting consistent utility and high level of demand and activity.

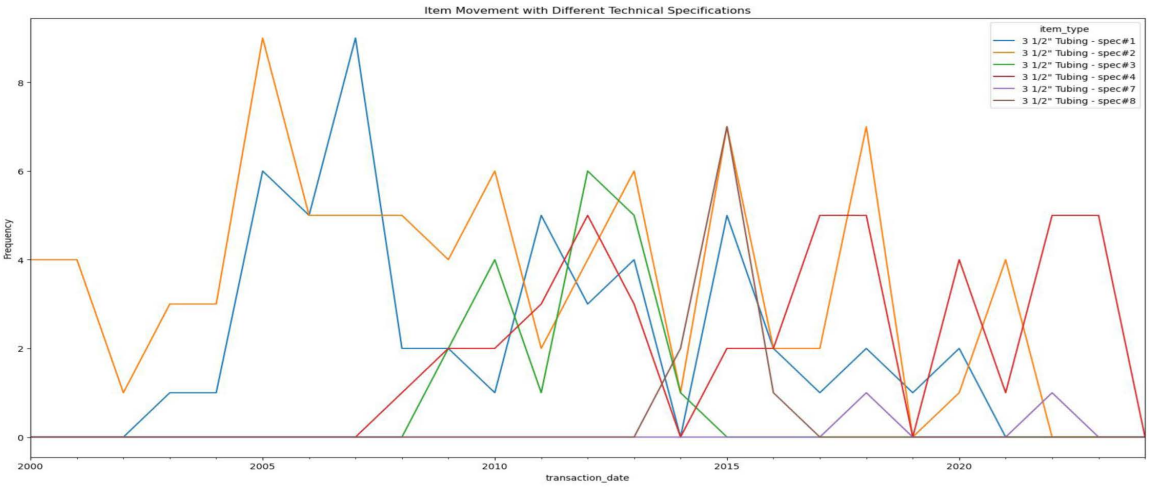

Figure 9—Overview of inventory status of the annual inventory movement for a sample of tubing SKUs with different technical specifications

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In (Fig. 10), performance and consistency of the inventory were assessed. Over 23 years, the inventory revealed more than 15,000 major transactions where issues accounted for 45% and returns represented almost 35% (Fig. 10a). It could be viewed from (Fig. 10b) that significant fluctuations have occurred in annual transactions, peaking at 125% in 2022, probably due to the increased oil prices and operations following the global recovery from the COVID-19 pandemic crisis in 2020 and 2021.

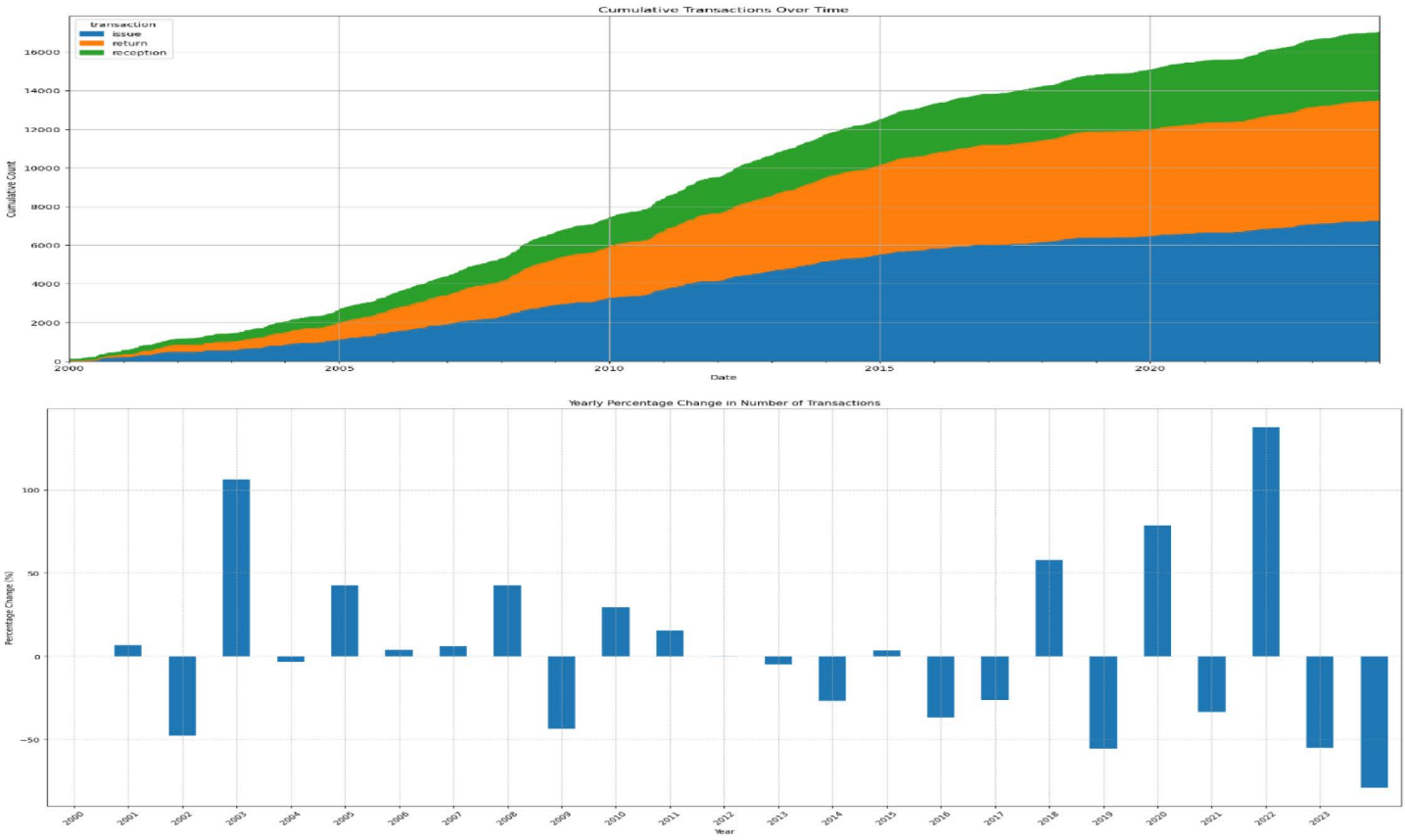

Figure 10—Inventory performance where a) a demonstration of cumulative total number of transactions, broken down and stacked by transaction type, and b) the annual variation in total number of transactions.

From a financial perspective, items that have not had any issuance transactions for a certain number of years are assumed outdated. When analyzed, it was revealed that those outdated SKUs follow the rule of 20/80 where only 20% of items account for more than 80% of value (Fig. 11). Further analytics assisted in defining which of these outdated items could be re-used again, sold, loaned, or junked. Coupled with other financial metrics, this could potentially result in a 10% savings of total inventor value.

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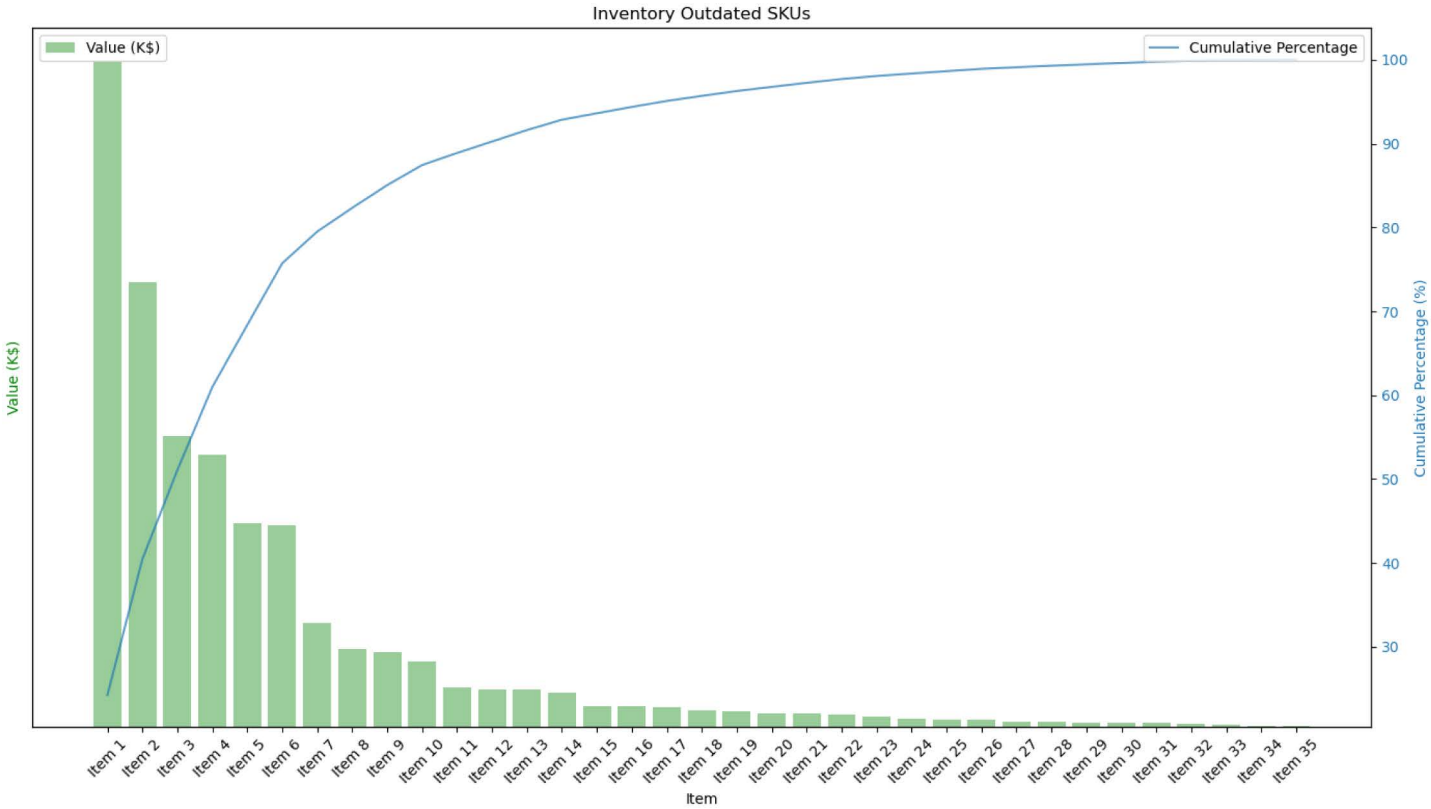

Figure 11—Pareto chart of outdated items with two axes of total value of items in K$ on the left axis, and cumulative percentages on the right axis

Limitations and Future Work

The presented synthetic dataset was generated according to authors' expertise and domain knowledge, which may introduce some subjectivity, despite efforts to minimize bias. To address this, the dataset was published as open source under MIT license. This approach should encourage engineers and supply chain professionals to contribute and refine the dataset, ensuring it would be representative of diverse scopes and experience.

Limitations

The type of unique SKUs and their detailed description in the data set are limited to common specification typically used in the MENA region. While the selection and creation of these descriptions were based on oil and gas wells, exceptional cases such as HPHT, multilateral, and smart completions were not accounted for.

The introduced work focused on data related to drilling and sidetrack activities but did not cover inventory management for workover and well intervention activities. In addition, the dataset emphasized the major items commonly associated with drilling activities but excluded rarely used items.

The analytics in this work were qualitative rather than quantitative. The primary objective of this paper was limited to producing the synthetic data set and presenting statistics to prove its similarity to real field inventories. While business analytics are crucial for identifying financial issues and optimizing drilling inventory, this was not the focus of this paper. Their brief mention in the discussion section was limited to illustrative purposes only.

Future Work

It is strongly recommended to further perform financial analysis of KPIs to identify key pain points in the introduced inventory management system. In addition, revealing diverse inventory metrics could highlight their essential role and ranking in cost savings. Some metrics are worth investigation on the addressed data set, which include comparing allocated and unallocated stock items, providing statistical insights

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into inventory dimensions, and identifying latent challenges such as stock count accuracy, scrap stock, underutilization of annual capital allocation, and excess inventory surplus.

In addition, it is necessary to investigate data driven analysis, which could be a key contributor to optimize inventory management. Comprehensive approaches of aging inventory, safety stock, service levels, and turnover ratios are necessary to be addressed, in order to optimize inventory policy, demand forecasting, and decision-making support.

As mentioned earlier, future contributions to the source of this open-source dataset are essential as a way forward to advance the potential for innovative approaches in drilling inventory management.

Conclusion

This paper presents a novel approach, which produces a synthetic dataset for drilling inventory systems, using a replicable iterative approach. Codebase was made open source to permit reproducibility and enable future innovative research on the constructed dataset. Financial analytics were briefly outlined, since they were a minor purpose of the research. To the authors' knowledge, this is the first open-source synthetic dataset for drilling inventory systems in the oil and gas industry.

The generated inventory system reflects real transactions, as thoroughly explained in the results and discussion sections. The introduced dataset consisted of nearly 500 unique SKUs, varying from drilling tubular to completion equipment, accessories, and production Xmas trees. Anomalous transactions, such as junk, sales, and loans were incorporated, while potential human and system errors were accounted for using random probabilities. A systematic approach was followed to estimate SKUs prices over time and estimate the quantities required for each drilled well.

This dataset is intended to serve as a benchmark for future research into the role of data analytics and machine learning in optimizing the performance of drilling inventories. This would allow the calculation of different inventory metrics, such as cash flow, in a drilling inventory system that simulates real data. Further research could lead to innovative approaches for inventory management, which could ultimately be applied to actual inventory data from oil companies, potentially resulting in significant cost savings in millions of dollars.

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